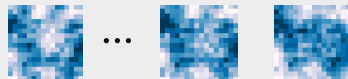


Input: Hydrometeorological Time Series

~8×8km context; 48 hour lookback

Precipitation



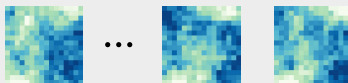
← Time

Soil Moisture



← Time

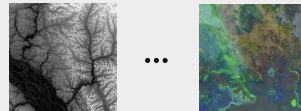
Snow Water Equivalent



← Time

Spatial Context: Terrain and AlphaEarth Conditioning - Same ~8×8km

[Height above nearest drainage, Topological Wetness Index, Global Curve Numbers (Dry, Average, Wet), POLARIS Soil Properties (ksat, clay, theta_s), Google AlphaEarth]



Conv2D Encoder



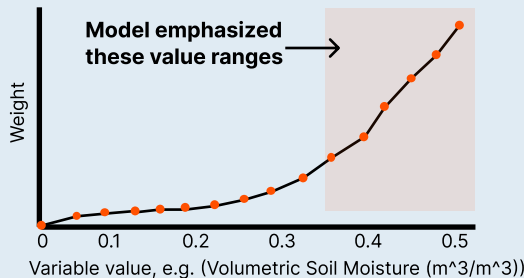
32 Parameters per variable

(16 control points: x_i, y_i)



Value Modulator

Learns a weighting function over each variable's raw value ranges. Runs variable-wise to normalize each variable to [0,1]



Conv2D Encoder

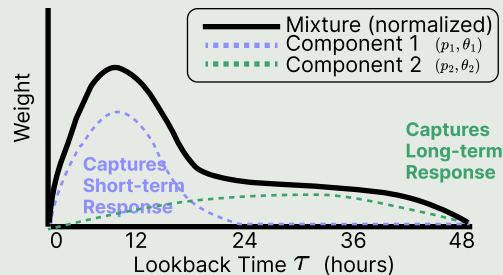


5 Parameters per variable ($p_1, \theta_1, p_2, \theta_2, w$)



Temporal Modulator

Learns a temporal lookback weighting function over the 48 hour look back window via a two-mixture Gamma kernel



Modulated output